

Project 0.1: Root Finding in One Dimension

1 Examples

The cases to be studied as examples are

$$F(x) \equiv 2x - 3 \sin x + 5 = 0 \quad (4)$$

and

$$F(x) \equiv x^3 - 8.5x^2 + 20x - 8 = 0 \quad (5a)$$

Note that equation (5a) can be factorised and rewritten as

$$F(x) \equiv (x - \frac{1}{2})(x - 4)^2 = 0 \quad (5b)$$

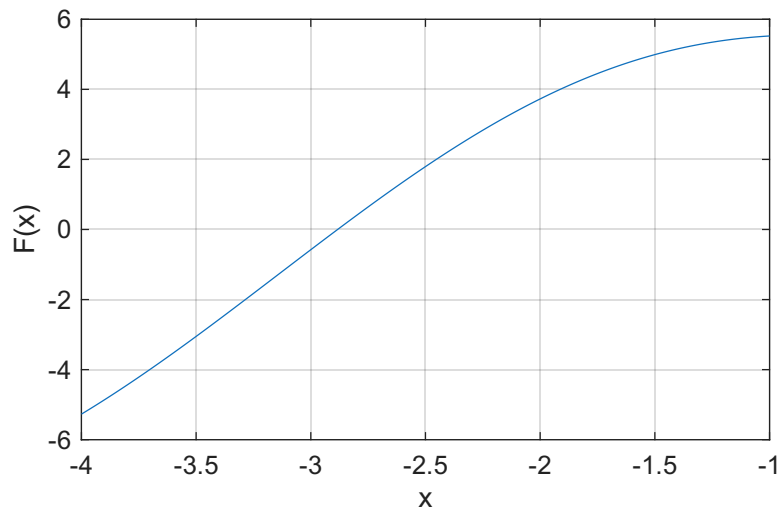
Question 1. Show, with the help of a graph, that equation (4) has exactly one root (which is in fact -2.88323687...).

Solution. Note that $-1 \leq \sin x \leq 1$ for all $x \in \mathbb{R}$. Thus,

$$x < -4 \implies F(x) = 2x - 3 \sin x + 5 < 0$$

$$x > -1 \implies f(x) = 2x - 3 \sin x + 5 > 0$$

and so the roots of equation (4) must lie in the range $-4 \leq x \leq -1$.



Sketching the graph of $F(x)$ for $-4 \leq x \leq -1$, we see that $F(x)$ is strictly increasing and thus one-to-one in the given domain, with $F(-4) < 0 < F(-1)$. Thus equation (4) has exactly one root, as required.

We can verify that $F(x)$ is strictly increasing by calculating $\frac{dF}{dx}$:

$$\frac{dF}{dx} = 2 - 3 \cos x$$

But $-4 \leq x \leq -1 \implies \cos x \leq \max(\cos(-4), \cos(-1)) < \frac{2}{3}$, and so $\frac{dF}{dx} > 0$ on the given domain. $\square$

2 Binary Search

Programming Task: write a program to solve equation (4) by binary search. Provide for termination of the iteration as soon as the truncation error is guaranteed to be less than 0.5×10^{-5} , and print out the number of iterations, N , as well as the estimate of the root. Run the program for a number of suitable starting values to check that it is working; include some of these results in your report.

Results. Note that a and b are the starting lower and upper bound for the binary search respectively, where we choose a, b s.t. $a \leq -4$ and $b \geq -1$ to ensure that $F(a)F(b) < 0$.

a	b	x_N	N
-4	-1	-2.8832	19
-7	0	-2.8832	20
-7.5	1.3	-2.8832	20
-100	100	-2.8832	25
-10000	10000	-2.8832	31

Question 2. Suppose that the rounding error in evaluating $F(x)$ in equation (4) is at most δ for $|x| < \pi$. By considering a Taylor expansion of $F(x)$ near x_* , or otherwise, estimate the accuracy that may be expected for the calculated value of the root.

Hint: Note that $|F'(x)| > 4$ for $-\frac{5}{4}\pi < x < -\frac{3}{4}\pi$.

Solution. We note that $F(-\pi) = -2\pi + 5 < 0$.

Also, $F(-\frac{3}{4}\pi) = -\frac{3}{2}\pi + \frac{3\sqrt{2}}{2} + 5 > 0$.

Thus, we can restrict our attention to the domain $x \in (-\pi, -\frac{3}{4}\pi)$, within which it is given that $|F'(x)| > 4$ and the maximum error in evaluating $F(x)$ is at most δ .

Taylor expanding around $x_* \in (-\pi, -\frac{3}{4}\pi)$:

$$F(x) = F(x_*) + (x - x_*)F'(x_*) + O[(x - x_*)^2]$$

But $F(x_*) = 0$, and for any calculated value of the root within the domain $(-\pi, -\frac{3}{4}\pi)$, we must have $|F(x) - 0| \leq \delta$. Thus, ignoring higher order terms, we have

$$|(x - x_*)F'(x_*)| \leq \delta$$

and since $F'(x_*) > 4$, this implies

$$|x - x_*| < \frac{\delta}{4}$$

□

3 Fixed-Point Iteration

There are many possible choices of f , e.g.

$$f(x) = x - h(F(x)) \quad (6)$$

for some function $h(F)$ such that $h(0) = 0$.

Programming Task: write a program to implement the iteration scheme in equation (1) for general f . Provide for termination of the process as soon as $|x_N - x_{N-1}| < \epsilon$ or when $N = N_{max}$, whichever occurs first. Print out the values of N and x_N for each N , so that you can watch the progress of the iteration.

Question 2. Use the program to solve (4) by fixed-point iteration by taking

$$h(F) = \frac{F}{2+k} \quad (7a)$$

in (6), so that

$$f(x) = \frac{3 \sin x + kx - 5}{2+k} \quad (7b)$$

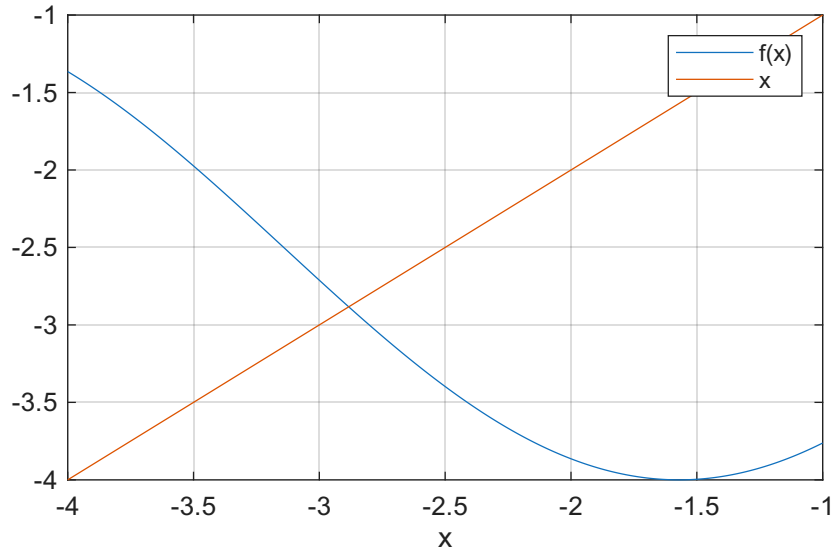
for some constant k .

- (i) First run the program with $k = 0$, $\epsilon = 10^{-5}$, $x_0 = -2$, $N_{max} = 10$. Plot $y = f(x)$ and $y = x$ on the same graph, and use these plots to show why convergence should not occur. Explain the divergence by identifying a theoretical criterion that has been violated.
- (ii) Determine the values of k for which convergence is guaranteed if x_N remains in the range $(-\pi, -\frac{1}{2}\pi)$.
- (iii) Choose, giving reasons, a value of k for which monotonic convergence should occur near the root, and also a value for which oscillatory convergence should occur near the root. Verify that these two values of k give the expected behaviour, by running the program with $N_{max} = 20$.
- (iv) Also run the case $k = 16$. This should converge only slowly, so set $N_{max} = 50$. Discuss whether the truncation error is expected to be less than 10^{-5} in this case?
- (v) Discuss whether your results are consistent with first-order convergence.

Solution.

(i) Program results.

N	x_N
0	-2
1	-3.8639
2	-1.5083
3	-3.9971
4	-1.3677
5	-3.9692
6	-1.3956
7	-3.9770
8	-1.3876
9	-3.9749
10	-1.3898



We see that we will reach a stable limit cycle of period 2, and hence the iteration scheme will not converge, but will oscillate between two distinct values on either side of the root.

From theory, if $x_* = f(x_*)$, f is continuously differentiable in some neighbourhood of x_* , and $|f'(x_*)| > 1$, then the sequence (x_N) defined by $x_N = f(x_{N-1})$ does not converge to x_* for any starting value $x_0 \neq x_*$.

¹ To see this, note that we can pick some interval $I = [x_* - \delta, x_* + \delta]$ and constant $\gamma > 1$ such that $f'(x) \geq \gamma \forall x \in I$. Suppose x_N lies in this interval for some N . Now, since $x_* = f(x_*)$ and $x_{N+1} = f(x_N)$, we have

$$|x_* - x_{N+1}| = |f(x_*) - f(x_N)| \geq \gamma |x_* - x_N|$$

¹Süli, E. and Mayers, D., *An Introduction to Numerical Analysis*, page 15.

where the last inequality follows from the Mean Value Theorem and the fact that $f'(x) > \gamma$ in I . Now if $x_{N+1} \in I$, then it follows that

$$|x_* - x_{N+2}| > \gamma^2 |x_* - x_N|$$

and so on for $x_{N+3}, x_{N+4}, \dots$. Since $\gamma > 1$, after finitely many steps, we can always find some k sufficiently large s.t. $x_{N+k} \notin I$, and so the sequence (x_N) does not converge to x_* .

Note that the root of equation (7b) is $x_* \in (-\pi, -\frac{3}{4}\pi)$. Also, in the region $x \in (-\pi, -\frac{3}{4}\pi)$, we have

$$f'(x) = \frac{3 \cos x}{2} \leq -\frac{3\sqrt{2}}{4} < -1$$

Thus $|f(x_*)| > 1$ and so the iteration scheme diverges.

- (ii) We know from theory that, if there exists an interval $[a, b]$ such that f is continuously differentiable, f has a unique fixed point $x_* \in [a, b]$, and

$$\max_{x \in [a, b]} |f'(x)| = \gamma < 1$$

then the iteration scheme will converge to the fixed point x_* for any initial guess $x_0 \in [a, b]$.² To see this, note that the requirement on $f'(x)$ implies (by the Mean Value Theorem) that

$$|f(x) - f(y)| \leq \gamma |x - y| \quad \forall x, y \in [a, b]$$

Then, since $x_* = f(x_*)$ and $x_N = f(x_{N-1})$, we have

$$|x_* - x_N| = |f(x_*) - f(x_{N-1})| \leq \gamma |x_* - x_{N-1}|$$

Let $\epsilon_N = |x_* - x_N|$. We see that $0 \leq \epsilon_N \leq \gamma^N \epsilon_0$, and since $\gamma < 1$, $\epsilon_N \rightarrow 0$ as $N \rightarrow \infty$, and thus $x_N \rightarrow x_*$, as required.

Note that $x_* \in (-\pi, -\frac{1}{2}\pi)$. Thus, for any $x_0 \in (-\pi, -\frac{1}{2}\pi)$, we can find some closed interval $[a, b] \subseteq (-\pi, -\frac{1}{2}\pi)$ such that $x_0, x_* \in [a, b]$. Then if $\max_{x \in [a, b]} |f'(x)| = \gamma$ for some $\gamma < 1$, by the above theorem, convergence is guaranteed. This is true if $-1 < f'(x) < 1 \quad \forall x \in (-\pi, -\frac{1}{2}\pi)$.

Now

$$f'(x) = \frac{3 \cos x + k}{2 + k}$$

For $k < -2$, we have

$$\frac{k}{2 + k} < f'(x) < \frac{k - 3}{2 + k}$$

We see that $\frac{k}{2+k} > 1$, so convergence is not guaranteed for any $k < -2$.

²Epperson, J.F., *An Introduction to Numerical Methods and Analysis*, page 130.

For $k > -2$, we have

$$\frac{k-3}{2+k} < f'(x) < \frac{k}{2+k}$$

We see that $\frac{k}{2+k} < 1$. Also, $\frac{k-3}{2+k} > -1 \implies k > \frac{1}{2}$. So we have guaranteed convergence for $k > \frac{1}{2}$. $\square$

- (iii) Suppose $|f'(x)| < 1$ on some region $[a, b]$ containing x_* , so convergence is guaranteed. By the mean value theorem,

$$x_n - x_* = f(x_{N-1}) - f(x_*) = f'(c_N)(x_{N-1} - x_*)$$

for some $c_N \in (x_{N-1}, x_N)$ (where, for the remainder of this project, we will take this to instead mean $c_N \in (x_N, x_{N-1})$ if $x_N < x_{N-1}$).

If we have $f'(x) > 0$ on $[a, b]$, then given $x_{N-1} \in [a, b]$, we see that

- (a) $|x_N - x_*| < |x_{N-1} - x_*|$, so in particular $x_N \in [a, b]$, and therefore
- (b) $c_N \in [a, b]$, so $f'(c_N) > 0$ and $\text{sgn}(x_N - x_*) = \text{sgn}(x_{N-1} - x_*)$, thus convergence is monotonic.

Conversely, if we have $f'(x) < 0$ on $[a, b]$, then given $x_{N-1} \in [a, b]$, we see that

- (a) $|x_N - x_*| < |x_{N-1} - x_*|$, so in particular $x_N \in [a, b]$, and therefore
- (b) $c_N \in [a, b]$, so $f'(c_N) < 0$ and $\text{sgn}(x_N - x_*) = -\text{sgn}(x_{N-1} - x_*)$, thus convergence is oscillatory.

Thus, near x_* , monotonic convergence should occur if $f'(x_*) > 0$ and oscillatory convergence should occur if $f'(x_*) < 0$.

From the discussion in part (ii), $k = 4 \implies f'(x) > \frac{1}{6} > 0$.

N	x_N	ϵ_N	$\epsilon_N/\epsilon_{N-1}$	$\text{sgn}(x_N - x_*)$
0	-2	0.88324		+1
1	-2.6213	0.26192	0.29655	+1
2	-2.8294	0.0538	0.2054	+1
3	-2.8732	0.010057	0.18693	+1
4	-2.8814	0.0018495	0.18391	+1
5	-2.8829	0.00033917	0.18338	+1
6	-2.8832	6.2161×10^{-5}	0.18328	+1
7	-2.8832	1.139×10^{-5}	0.18323	+1
8	-2.8832	2.0852×10^{-6}	0.18308	+1

So for $k = 4$, we get monotonic convergence near the root, as expected.

For $k = \frac{1}{2}$, since $x_* < -\frac{3}{4}\pi$ and $f'(x) = \frac{3\cos x + k}{2+k}$, we see that $f'(x_*) < f'(-\frac{3}{4}\pi) = \frac{-3\sqrt{2}+1}{5} < 0$.

N	x_N	ϵ_N	$\epsilon_N/\epsilon_{N-1}$	$\text{sgn}(x_N - x_*)$
0	-2	0.88324		+1
1	-3.4912	0.60792	0.68829	-1
2	-2.2872	0.59599	0.98038	+1
3	-3.3624	0.47918	0.80401	-1
4	-2.4096	0.4736	0.98834	+1
5	-3.2839	0.40068	0.84603	-1
6	-2.4866	0.39667	0.98998	+1
7	-3.2283	0.34509	0.86997	-1
8	-2.5417	0.34152	0.98966	+1
9	-3.1858	0.30256	0.88591	-1
10	-2.5841	0.2991	0.98858	+1
11	-3.1517	0.26843	0.89744	-1
12	-2.6182	0.26499	0.98719	+1
13	-3.1234	0.24015	0.90626	-1
14	-2.6465	0.23671	0.98569	+1
15	-3.0994	0.21618	0.91325	-1
16	-2.6705	0.21276	0.98417	+1
17	-3.0787	0.19551	0.91895	-1
18	-2.6911	0.19212	0.98267	+1
19	-3.0607	0.17746	0.92369	-1
20	-2.7091	0.17413	0.98123	+1

So for $k = 1/2$, we get oscillatory convergence near the root, as expected.

□

(iv) Program results.

N	x_N	ϵ_N	$\epsilon_N/\epsilon_{N-1}$
0	-2	0.88324	
1	-2.2071	0.67613	0.76552
2	-2.3737	0.50954	0.75361
3	-2.5035	0.37973	0.74525
4	-2.6024	0.28085	0.73959
5	-2.6766	0.20665	0.7358
6	-2.7317	0.15153	0.73328
7	-2.7724	0.11086	0.73159
8	-2.8023	0.080976	0.73044
9	-2.8242	0.059084	0.72965
10	-2.8402	0.043079	0.7291
11	-2.8518	0.031392	0.72872
12	-2.8604	0.022868	0.72845
13	-2.8666	0.016653	0.72825
14	-2.8711	0.012126	0.72812
15	-2.8744	0.0088277	0.72802
16	-2.8768	0.006426	0.72794
17	-2.8786	0.0046775	0.72789
18	-2.8798	0.0034045	0.72785
19	-2.8808	0.0024779	0.72783
20	-2.8814	0.0018034	0.72781
21	-2.8819	0.0013125	0.72779
22	-2.8823	0.00095523	0.72778
23	-2.8825	0.00069519	0.72777
24	-2.8827	0.00050594	0.72777
25	-2.8829	0.0003682	0.72776
26	-2.883	0.00026796	0.72776
27	-2.883	0.00019501	0.72776
28	-2.8831	0.00014192	0.72775
29	-2.8831	0.00010328	0.72775
30	-2.8832	7.5164e-05	0.72775
31	-2.8832	5.47e-05	0.72775
32	-2.8832	3.9808e-05	0.72774
33	-2.8832	2.8969e-05	0.72774
34	-2.8832	2.1082e-05	0.72773

We see that, by our calculation of the derivative in part (ii),

$$\frac{13}{18} < f'(x) < \frac{8}{9}$$

for all $x \in (-\pi, -\frac{1}{2}\pi)$. By the mean value theorem,

$$|f(x) - f(y)| = \alpha|x - y|$$

for all $x, y \in (-\pi, -\frac{1}{2}\pi)$, where $\alpha \in (\frac{13}{18}, \frac{8}{9})$.

Since $x_* = f(x_*)$ and $x_N = f(x_{N-1})$, and writing $\alpha = \gamma_N$, we have

$$|x_* - x_N| = |f(x_*) - f(x_{N-1})| = \gamma_N |x_* - x_{N-1}|$$

Rewriting this in terms of ϵ_N and ϵ_{N-1} :

$$\epsilon_N = \gamma_N \epsilon_{N-1} \text{ for some } \gamma_N \in \left(\frac{13}{18}, \frac{8}{9}\right)$$

In particular, $\gamma_N > \frac{1}{2}$, so $\epsilon_N > \frac{1}{2} \epsilon_{N-1}$, and therefore

$$\begin{aligned} \epsilon_N &> \epsilon_{N-1} - \epsilon_N \\ &= |x_{N-1} - x_*| - |x_N - x_*| \\ &= |x_{N-1} - x_N| \end{aligned}$$

where the last equality follows from the fact that $f'(x_*) \geq \frac{13}{18} > 0$, so convergence is monotonic and $\text{sgn}(x_{N-1} - x_*) = \text{sgn}(x_N - x_*)$.

Thus, the truncation error $\epsilon_N = |x_N - x_*|$ is expected to be greater than the termination threshold $\epsilon = 10^{-5} < |x_N - x_{N-1}|$, which is reflected by the error values above. $\square$

- (v) From the proof of the theorem in part (i), we see that since $\epsilon_N < \gamma^N \epsilon_0$ and $\gamma < 1$, the iteration method is indeed first-order convergent whenever it converges, as stated in the given background information. This agrees with our error estimates in part (iii)-(iv) (which were calculated by taking $x_* = -2.88323687$), as the value of $\epsilon_N/\epsilon_{N-1}$ appears to converge to a constant in all three cases investigated. $\square$

Question 4. Now use your program to find the *double root* of equation (5a) by taking

$$h(F) = \frac{1}{20}F \tag{8a}$$

in (6), so that

$$f(x) = \frac{1}{20}(-x^3 + 8.5x^2 + 8) \tag{8b}$$

By considering $f'(x_*)$, explain why convergence will be slow at a multiple root for any choice of differentiable function h in (6).

In your calculations some care may be needed over the choice of x_0 . Also,

- (a) since convergence will be slow, take $N_{max} = 1000$;
- (b) suppress the printing of each iterate, but print out the *final* values of N and x_N .

Is this an example of first-order convergence? Does the termination criterion ensure a truncation error of less than 10^{-5} ?

Note: it can be shown that the truncation error ϵ_N is asymptotic to $\frac{40}{7N}$ as $N \rightarrow \infty$.

Solution. From equation (5b), the double root occurs at $x_* = 4$. Since $F'(x_*) = 0$ at a double root,

$$f'(x_*) = 1 - h'(F(x_*))F'(x_*) = 1$$

for any choice of $h(F)$.

We see that since $f'(x)$ is continuous, as $x \rightarrow x_*$, we have $f'(x) \rightarrow 1$. Then, by the mean value theorem, $\frac{\epsilon_N}{\epsilon_{N-1}} = |f'(c_N)|$, where $c_N \in (x_{N-1}, x_N)$. But c_N converges to x_* if $x_N \rightarrow x_*$, so $\frac{\epsilon_N}{\epsilon_{N-1}} \rightarrow |f(x_*)| = 1$. Therefore, the truncation error decays slowly and convergence will be slow.

Program results.

x_0	x_N	N	ϵ_N
-3	4.0075	742	0.0075324
-2.55	3.9926	2	0.0073782
4.1	4.0075	694	0.0075394
4.5	4.0075	736	0.0075349
5	4.0075	740	0.0075342
6	4.0075	740	0.0075406

Given that the truncation error ϵ_N is asymptotic to $\frac{40}{7N}$ as $N \rightarrow \infty$, we must have

$$\lim_{N \rightarrow \infty} \frac{\epsilon_N}{\epsilon_{N-1}} = 1$$

Thus, this is not an example of first-order convergence.

Since $\frac{\epsilon_N}{\epsilon_{N-1}} \rightarrow 1$, we have $\frac{\epsilon_N}{\epsilon_{N-1}} > \frac{1}{2}$ for N sufficiently large, and thus, by our earlier discussion in Question 3 (iv), the termination criterion does not ensure a truncation error of less than 10^{-5} .

4 Newton-Raphson Iteration

A refinement of (6) is to let h depend on the derivatives of F , i.e.

$$f(x) = x - h(F, F', F'', \dots) \quad (9a)$$

In Newton-Raphson iteration,

$$h = \frac{F}{F'} \quad (9b)$$

Programming Task: modify your program to recalculate the root of equation (4), and the double root of equation (5a), using Newton-Raphson iteration.

Question 5. For equation (4), experiment with various x_0 until you have demonstrated a case that converges, and also a case that has not converged in 10 iterations. In the unconverged case, show graphically what happened in the first few iterations.

For both equation (4) and equation (5a) do your (converged) results bear out the theoretical orders of convergence? Comment on the effects of rounding error.

Hint: you may want to use a smaller value for ϵ .

Solution. For equation (4), $h(x) = \frac{2x-3\sin x+5}{2-3\cos x}$, so

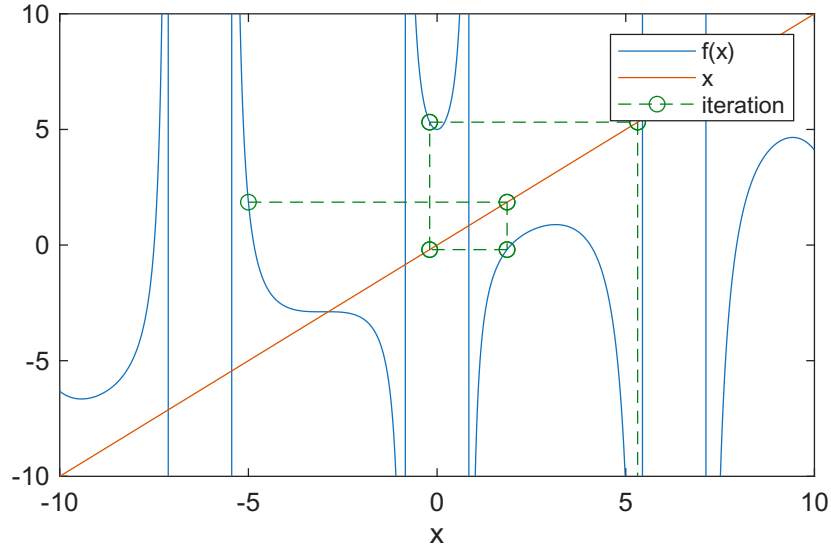
$$f(x) = \frac{-3x \cos x + 3 \sin x - 5}{2 - 3 \cos x}$$

We set $\epsilon = 10^{-10}$ and $N_{max} = 10$.

x_0	x_N	N
-2	-2.883236873	5
-3	-2.883236873	4
-4.5	-2.883236873	6
-5	-284.209361	10

So Newton-Raphson iteration converges for $x_0 = -2, -3, -4$ and does not converge within 10 iterations for $x_0 = -5$. We look further into the behaviour of the $x_0 = -5$ case in its first few iterations.

N	x_N
0	-5
1	1.85524862
2	-0.1965688965
3	5.314604572
4	-54.90496787
5	-6.313063734
6	-13.85967402
7	2.975089553
8	0.8670021682
9	-74.94616944
10	-284.209361



Plotting the first four iterations, we see that the $x_0 = 5$ case does not converge within 10 iterations because $f(x)$ diverges whenever $2 - 3 \cos x = 0$.

For equation (5a), $h(x) = \frac{x^3 - 8.5x^2 + 20x - 8}{3x^2 - 17x + 20}$, so

$$f(x) = \frac{2x^3 - 8.5x^2 + 8}{3x^2 - 17x + 20}$$

We set $\epsilon = 10^{-7}$ and $N_{max} = 100$ as convergence is slow. Program results.

x_0	x_N	N
2.5	3.999999922	23
3	3.999999933	23
5	4.000000082	24
20	4.000000141	29

For equation (4), we expect the Newton-Raphson method to converge quadratically for appropriate x_0 .³ Note that for $x \in [-3, -\frac{5}{2}] \subseteq (-\pi, -\frac{1}{2}\pi)$, we have

$$|F'(x)| = |2 - 3 \cos(x)| \geq 2$$

$$|F''(x)| = |3 \sin(x)| \leq 3$$

Thus, $\frac{|F''(y)|}{|F'(x)|} \leq \frac{3}{2}$ for all $x, y \in [-3, -\frac{5}{2}]$.

Now suppose $x_N \in [-3, -\frac{5}{2}]$. Then, by Taylor's theorem (expanding about $x = x_N$),

³Süli, E. and Mayers, D., *An Introduction to Numerical Analysis*, page 23.

$$0 = F(x_*) = F(x_N) + (x_* - x_N)F'(x_N) + \frac{(x_* - x_N)^2}{2}F''(c_N)$$

for some $c_N \in (x_*, x_N) \subseteq [-3, -\frac{5}{2}]$.

Rearranging, we see that

$$\begin{aligned} x_* - x_{N+1} &= x_* - x_N + \frac{F(x_N)}{F'(x_{N-1})} \\ &= -\frac{(x_* - x_N)^2}{2F'(x_N)}F''(c_N) \end{aligned}$$

Therefore,

$$\begin{aligned} \epsilon_{N+1} &= |x_* - x_{N+1}| \\ &= |x_* - x_N|^2 \frac{F''(c_N)}{2F'(x_N)} \\ &\leq \frac{3}{4}\epsilon_N^2 \end{aligned}$$

This tells us that, if $x_0 \in [-3, -\frac{5}{2}]$ (and thus $\epsilon_0 < \frac{1}{2}$), the Newton-Raphson method will converge at least quadratically.

Taking $x_* = -2.8832368725582835367$, we run the program for 6 iterations with $x_0 = -2.5$.

N	x_N	ϵ_N	$\epsilon_N/\epsilon_{N-1}^2$
0	-2.5	0.38324	
1	-2.907731265651990249	0.024494	0.16678
2	-2.8832807269653808468	4.3854×10^{-5}	0.073094
3	-2.8832368727086699067	1.5039×10^{-10}	0.078195
4	-2.8832368725582835367	1.4967×10^{-18}	66.1807
5	-2.8832368725582835367	1.4985×10^{-18}	$6.689061675879693 \times 10^{17}$

We see that our results for the first three iterations appear to be consistent with quadratic convergence, but this breaks down for iterations $N = 4, 5$. This is likely due to insufficient precision in the value of x_* used, as well as rounding errors in the calculation of $f(x_N)$ and ϵ_N .

For equation (5a), due to the double root, we expect the Newton-Raphson method to converge linearly for appropriate x_0 .⁴ Note that for $x \in [\frac{15}{4}, \frac{17}{4}] \setminus \{4\}$, we have

$$F'(x) = 3x^2 - 17x + 20 \neq 0$$

$$|F''(x)| = |6x - 17| \in [\frac{11}{2}, \frac{17}{2}]$$

⁴Süli, E. and Mayers, D., *An Introduction to Numerical Analysis*, page 36.

Thus, $\frac{|F''(y)|}{|F''(x)|} \leq \frac{17}{11}$ for all $x, y \in [\frac{15}{4}, \frac{17}{4}]$.

Suppose $x_N \in [\frac{15}{4}, \frac{17}{4}] \setminus \{4\}$. By Taylor's theorem (expanding about $x = x_N$),

$$0 = F(x_*) = F(x_N) + (x_* - x_N)F'(x_N) + \frac{(x_* - x_N)^2}{2}F''(c_N)$$

for some $c_N \in (x_*, x_N) \subseteq [\frac{15}{4}, \frac{17}{4}]$.

Rearranging, we see that

$$\begin{aligned} x_* - x_{N+1} &= x_* - x_N + \frac{F(x_N)}{F'(x_{N-1})} \\ &= -\frac{(x_* - x_N)^2}{2F'(x_N)}F''(c_N) \end{aligned}$$

By the mean value theorem applied to F on (x_*, x_N) , we know that

$$\begin{aligned} F'(x_N) &= F'(x_N) - F'(x_*) \\ &= (x_N - x_*)F''(d_N) \end{aligned}$$

for some $d_N \in (x_*, x_N)$.

Substituting this into our previous equation yields

$$x_* - x_{N+1} = (x_* - x_N) \frac{F''(c_N)}{2F''(d_N)}$$

Therefore,

$$\begin{aligned} \epsilon_{N+1} &= |x_* - x_{N+1}| \\ &= |x_* - x_N| \frac{|F''(c_N)|}{|2F''(d_N)|} \\ &= \frac{1}{2} \epsilon_N \frac{|F''(c_N)|}{|F''(d_N)|} \\ &\leq \frac{17}{22} \epsilon_N \end{aligned}$$

From the last inequality, it follows that $\epsilon_N \rightarrow 0$, so the Newton-Raphson method converges if $x_0 \in [\frac{15}{4}, \frac{17}{4}]$. Also, since F'' is continuous, and $c_N, d_N \in (x_*, x_N)$, it follows that as $N \rightarrow \infty$, $x_N \rightarrow x_*$, so $F''(c_N), F''(d_N) \rightarrow F''(x_*) = 7 \neq 0$. Then $\frac{F''(c_N)}{F''(d_N)} \rightarrow 1$, and so $\frac{\epsilon_{N+1}}{\epsilon_N} \rightarrow \frac{1}{2}$. This means that the Newton-Raphson method converges linearly with $C = \frac{1}{2}$.

We run the program for 10 iterations with $x_0 = \frac{17}{4}$.

N	x_N	ϵ_N	$\epsilon_N/\epsilon_{N-1}$
0	4.25	0.25	
1	4.12903	0.12903226	0.51613
2	4.06564	0.065643048	0.50873
3	4.03312	0.033120889	0.50456
4	4.01664	0.016637704	0.50233
5	4.00834	0.0083384845	0.50118
6	4.00417	0.004174191	0.50059
7	4.00209	0.0020883379	0.5003
8	4.00104	0.0010444802	0.50015
9	4.00052	0.00052231797	0.50007
10	4.00026	0.00026117846	0.50004

This is consistent with linear convergence with $C = \frac{1}{2}$.

5 Program Listings

Question 1: Binary Search

```

lb = -10000 % lower bound
ub = 10000 % upper bound
N = 0 % iteration number
tError = (ub - lb)/2 % size of possible truncation error
x = (ub+lb)/2 % starting value = midpoint
while tError >= 0.5 * 10^(-5)
    N = N+1 % increment N
    if 2*x - 3*sin(x) + 5 == 0 % break if we landed on the root
        break
    elseif 2*x - 3*sin(x) + 5 > 0
        ub = x % pick the left half-interval
    elseif 2*x - 3*sin(x) + 5 < 0
        lb = x % pick the right half-interval
    end
    x = (ub+lb)/2 % update midpoint
    tError = (ub-lb)/2 % update max possible error
end
disp(['x = ' num2str(x) ', N = ' num2str(N) ', tError = ' num2str(tError)])

```

Questions 3 & 4: Fixed-Point Iteration

```

x = 4.5; % set initial value x_0
eps = 10^(-5); % set termination error
Nmax = 1000; % set max number of iterations
N = 0; % initialise number of iterations to zero

errN = abs(4-x); %initialise Nth term error
errNm = abs(4-x); %initialise N-1th term error

while N < Nmax
    N=N+1; % increment N
    y = f(x); % set y = f(x_N-1)
    errNm = errN; % update errNm
    errN = abs(4-y); % update errN
    if abs(y-x) < eps % break once required accuracy reached
        break;
    end
    x = y; % set x_N = y
end

disp(['N = ' num2str(N) ', x_N = ' num2str(y) ...
      ', ' num2str(errN) ' & ' num2str(errN/errNm)])

```

Question 5: Newton-Raphson Iteration

```

x = 4.25; % set initial value x_0
eps = 0; % set termination error
Nmax = 10; % set max number of iterations
N = 0; % initialise number of iterations to zero
errN = abs(4-x); %initialise Nth term error
errNm = abs(4-x); %initialise N-1th term error

disp([num2str(N) ' & ' num2str(x) ' & ' num2str(errN) ' & '\\'])
while N < Nmax
    N=N+1; % increment N
    y = h(x); % set y = f(x_N-1)
    errNm=errN; % update errNm
    errN = abs(4-y); % update errN
    disp([num2str(N) ' & ' num2str(y,6) ' & ' num2str(errN, 8) ...
          ' & ' num2str(errN/(errNm)) ' '\\']) %print results in LaTeX table
    if abs(y-x) < eps % break once required accuracy reached
        break;
    end
    x = y; % set x_N = y
end

```